

YI-CHI (TRISTA) CHEN

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EDUCATION

San José State University

B.S. in Software Engineering

San Jose, CA

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Database Systems, Machine Learning for Big Data, Operating Systems

EXPERIENCE

Associate Test Technician

June 2026 – Present

Element Materials Technology

Morgan Hill, CA

- Perform **EMC/RF testing and validation** across **Wi-Fi 2.4/5/6E GHz**, Bluetooth, UWB, and IEEE 802.15.4 wireless technologies
- Measure peak and average RF and tune **transmit power/EIRP** using **spectrum analyzers**, call boxes, and antenna systems
- Investigate RF across bands, orientations, and configurations to identify anomalies for **certification** evaluations

Software Engineer – Mission Control Team

Jan 2026 – Aug 2026

SJSU Robotics Club

San Jose, CA

- Integrated **React + Python** robotic-arm control with **CAN communication** and feedback visualization
- Implemented command parsing, **ACK handling**, and **recovery logic** for reliable hardware interaction
- Built slider/gamepad controls with **joint limits**, deadzone/disconnect handling, and commanded-vs-feedback TX indicators

Software Engineer Intern

Mar 2026 – May 2026

SJSU Software and Computer Engineering Society

San Jose, CA

- Built **SCEvents**, a high-concurrency registration platform using **Go, Kafka, Redis, and MongoDB** to support 1k+ concurrent users
- Engineered **Kafka-based event-driven architecture** (producer-consumer pipelines) to process registrations and waitlist flows
- Integrated **Redis locks and caching** to prevent race conditions and ensure consistency under k6-simulated spikes of 10,000+ writes
- Built **React** calendar UI with **JWT-authenticated registration**, role-based visibility, and reusable components

Firmware Software Engineer Intern

April 2026 – May 2026

Alef

San Mateo, CA

- Built an **ESP32-based wheel calibration system** using dual Hall sensors to restore zero-position after power loss
- Implemented real-time motor control over **CAN (TWAI)**, including velocity, incremental positioning, and zero persistence
- Designed Hall-sensor signal processing and automated homing logic with hysteresis for reliable alignment under noise
- Exposed motor control and calibration through a lightweight **web interface** for real-time diagnostics and testing

Machine Learning Engineer

Feb 2026 – May 2026

SJSU AI/ML Club (Industry-Led Project, Nuvoton)

San Jose, CA

- Trained **YOLOv8** models for real-time headcount estimation, achieving **mAP@0.5 0.979** and **mAP@0.5:0.95 0.651**
- Ran 20–200+ epoch experiments, resolving dataset/configuration issues to improve accuracy by **>30%**
- Deployed models on edge hardware (Ethos-U55), maintaining **20–22 FPS** real-time inference

PROJECTS

Customer Churn Prediction | Python, scikit-learn, FastAPI

- Cleaned telecom data and engineered usage, tenure, and service-based features for churn prediction.
- Evaluated 3 classification models; deployed the **79%-recall logistic-regression pipeline** through FastAPI.
- Saved preprocessing, encoding, scaling, and inference in a reusable **scikit-learn pipeline**.

AgentDex | Python, RedisVL, uAgents, Playwright

- Built a multi-agent pipeline that crawls Wikipedia, structures research results, and stores them in Redis.
- Combined semantic KNN cache search with LLM-filtered prefetching to serve predicted follow-up queries faster.
- Implemented exact and semantic **Redis cache checks** to return warm results without re-crawling.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Go, SQL, JavaScript, TypeScript, HTML/CSS

Backend, Cloud & Data: Kafka, Redis, MongoDB, MySQL, RESTful APIs, FastAPI, AWS, Docker, GitHub Actions, CI/CD ,

Frontend, ML & Tools: React, scikit-learn, YOLOv8, Git, Linux, k6, Postman, JMeter, Playwright, CAN